COMPUTER SOFTWARE COLLEGE

An ISO 9001 : 2015 Certified Institution

PowerBI Project

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CHAPTER 1 – PROJECT OVERVIEW

1.1 Project Title

HR Recruitment Analytics Dashboard Using MySQL and Power BI

1.2 Business Scenario

Recruitment is one of the most critical business functions within an organization. Companies continuously hire candidates across multiple departments, locations, job roles, and experience levels. Managing recruitment activities manually often results in delayed hiring decisions, poor visibility into recruitment performance, inefficient recruiter utilization, and increased hiring costs.

Human Resource teams require a centralized analytics solution capable of tracking recruitment activities from application stage to candidate joining. Stakeholders need visibility into application trends, hiring funnels, recruiter productivity, hiring costs, candidate quality, sourcing effectiveness, and overall recruitment efficiency.

To address these challenges, this project focuses on building a comprehensive HR Recruitment Analytics Dashboard using MySQL and Power BI. The solution transforms raw recruitment data into meaningful business insights through dimensional modeling, DAX calculations, and interactive dashboards.

1.3 Project Objective

The primary objective of this project is to design and develop a complete business intelligence solution that converts recruitment data into actionable insights.

The project aims to:

• Design a structured Star Schema data model

• Create fact and dimension tables for efficient reporting

• Analyze recruitment performance across departments

• Track candidate progression through hiring stages

• Evaluate recruiter productivity

• Measure hiring costs and recruitment efficiency

• Assess candidate quality through score analysis

• Implement DAX calculations for KPI reporting

• Develop interactive Power BI dashboards

• Support data-driven recruitment decisions

1.4 Expected Outcome

Upon successful completion of the project, stakeholders should be able to:

• Monitor total applications and hiring trends

• Analyze department-wise recruitment performance

• Evaluate recruiter effectiveness

• Track offer acceptance rates

• Monitor average hiring costs

• Measure time-to-fill and time-to-join metrics

• Analyze candidate quality scores

• Compare recruitment sources

• Identify high-performing recruiters

• Improve recruitment planning and workforce management

The final deliverable consists of two interactive dashboard pages:

Executive Dashboard

Focused on recruitment performance monitoring.

Candidate Quality Dashboard

Focused on candidate quality and recruiter effectiveness.

1.5 Scope of Analysis

The scope of this project includes:

Recruitment Analysis

- Total Applications
- Total Offers
- Total Hires
- Offer Acceptance Rate
- Candidate Funnel Analysis

Recruiter Analysis

- Recruiter Productivity
- Hiring Success Rate
- Candidate Quality Assessment

Department Analysis

- Department-wise Hiring
- Hiring Cost Comparison
- Department Performance

Candidate Analysis

- Experience Analysis
- Education Analysis
- Age Analysis
- Candidate Quality Metrics

Source Analysis

- Referral Analysis
- Job Portal Analysis
- Recruitment Source Effectiveness

Operational Analysis

- Time To Fill
- Time To Join
- Offer Conversion

• Hiring Cost Monitoring

1.6 Project Deliverables

SQL Deliverables

• Database Creation Scripts

• Raw Table Creation

• Dimension Table Creation

• Fact Table Creation

• Star Schema Implementation

• Relationship Mapping

Power BI Deliverables

• Power BI Data Model

• DAX Measures

• Calculated Columns

• KPI Cards

• Interactive Dashboards

• Conditional Formatting

• Tooltips

• Drill Down Functionality

Reporting Deliverables

• Executive Dashboard

• Candidate Quality Dashboard

• Business Insights Report

• Project Documentation

1.7 Technology Stack

Database : MySQL

Data Modeling : Star Schema

ETL : SQL

Visualization : Power BI

Analytics Language : DAX

Reporting Tool : Power BI Desktop

Dataset : HR Recruitment Analytics Dataset

1.8 Success Criteria

The project will be considered successful when:

• All data is successfully transformed into a Star Schema

• Dashboard KPIs produce accurate results

• Interactive filtering works correctly

• Drill-down functionality operates successfully

• Conditional formatting behaves correctly

• Dashboard performance remains responsive

• Business users can derive insights without SQL knowledge

• Recruitment teams can identify trends and bottlenecks quickly

CHAPTER 2 – DATASET OVERVIEW

2.1 Dataset Description

The dataset represents recruitment activities performed across multiple departments, locations, recruiters, and job roles.

Each row represents a candidate application and contains details regarding application status, interview performance, offer information, joining information, compensation details, candidate scores, recruiter details, and department information.

The dataset was designed to simulate a real-world enterprise recruitment environment and provides sufficient volume for dimensional modeling and dashboard development.

The dataset supports:

• Recruitment Performance Analysis

• Candidate Quality Analysis

• Hiring Cost Analysis

• Recruiter Productivity Analysis

• Department Performance Analysis

• Source Effectiveness Analysis

2.2 Source System

Source Type : Recruitment Dataset

Storage Format : CSV

Loading Tool : MySQL Workbench

Reporting Tool : Power BI Desktop

Modeling Technique : Star Schema

2.3 Dataset Statistics

Total Records : 2,000

Total Columns : 35

Departments : 10

Recruiters : 25

Locations : 7

Candidate Sources : 6

Analysis Period : 2 Years

Distinct Candidates : 2,000

2.4 Dataset Structure

Each row represents a candidate recruitment event.

Example

Application ID : APP0001

Candidate : Candidate A

Department : Data Engineering

Recruiter : Recruiter 01

Source : LinkedIn

Application Status : Applied

2.5 Data Dictionary

Candidate_ID

Unique identifier for candidate

Candidate_Name

Candidate name

Age

Candidate age

Gender

Candidate gender

Education

Educational qualification

Experience_Years

Years of experience

Department

Hiring department

Recruiter_Name

Assigned recruiter

Location

Hiring location

Application_Date

Date application received

Interview_Date

Date interview conducted

Offer_Date

Date offer released

Joining_Date

Date candidate joined

Skill_Score

Skill evaluation score

Technical_Score

Technical evaluation score

Communication_Score

Communication assessment score

Overall_Rating

Overall candidate rating

Current_CTC_LPA

Current compensation

Expected_CTC_LPA

Expected compensation

Offered_CTC_LPA

Offered compensation

Hiring_Cost

Recruitment cost incurred

Time_To_Fill_Days

Time required to fill position

Time_To_Join_Days

Time required to join

CHAPTER 3 – DATA ARCHITECTURE

3.1 Architecture Overview

The project follows a three-layer architecture.

CSV File

↓

Raw Layer

↓

Star Schema

↓

Power BI Dashboard

The architecture ensures scalability, maintainability, and reporting efficiency.

3.2 Raw Layer

Table Name

recruitment_raw

Purpose

• Preserve original source data

• Support reprocessing

• Act as foundation for dimensional modeling

Characteristics

Records : 2,000

Columns : 35

Granularity : Candidate Level

3.3 Dimensional Modeling Approach

A Star Schema architecture was selected.

Reasons

• Better reporting performance

• Easier DAX calculations

• Industry-standard BI design

• Simpler business understanding

3.4 Fact Table Design

Fact Table

fact_recruitment

Purpose

Store recruitment transactions and measurable business events.

Grain

One row represents one candidate application.

Measures

• Hiring Cost

• Skill Score

• Technical Score

• Communication Score

• Overall Rating

• Offered CTC

• Time To Fill

• Time To Join

3.5 Dimension Tables

dim_candidate

Stores candidate information

dim_department

Stores department details

dim_recruiter

Stores recruiter information

dim_location

Stores location information

dim_source

Stores candidate source information

dim_date

Stores date attributes

3.6 Star Schema Architecture

dim_date

|

dim_candidate ---- fact_recruitment ---- dim_department

|

dim_recruiter

|

dim_location

|

dim_source

3.7 Relationship Design

Relationship Type

One-to-Many

Cross Filter Direction

Single

Active Relationships

Enabled

All dimensions connect directly to the fact table.

3.8 Architecture Benefits

- Improved Power BI performance
- Simplified reporting
- Easier maintenance
- Scalable architecture

CHAPTER 4 – DATA MODELING APPROACH

4.1 Introduction

Data modeling is one of the most important phases in Business Intelligence projects. A properly designed model improves report performance, simplifies calculations, and enables scalable reporting.

For this project, a Star Schema data model was implemented to support efficient recruitment analytics reporting.

4.2 Why Star Schema?

The following modeling techniques were evaluated:

Flat Table Model

Advantages

• Easy to understand

Disadvantages

• Poor performance

• Data redundancy

• Difficult maintenance

Snowflake Schema

Advantages

• Highly normalized

Disadvantages

• Complex relationships

• Increased query complexity

Star Schema

Advantages

• Fast reporting

• Simplified relationships

• Easy DAX calculations

• Industry standard

• Better Power BI performance

For these reasons, Star Schema was selected.

4.3 Fact Table Design

Fact Table Name

fact_recruitment

Purpose

Store recruitment transactions and measurable business metrics.

Grain

One record represents one candidate application.

Primary Key

application_id

Foreign Keys

candidate_id
department_id
recruiter_id
location_id
source_id
application_date_key

4.4 Fact Table Measures

Hiring Cost

Represents recruitment expenditure incurred for a candidate.

Offered CTC

Final compensation offered to candidate.

Skill Score

Technical skill evaluation score.

Communication Score

Communication assessment score.

Technical Score

Technical interview score.

Overall Rating

Overall candidate assessment score.

Time To Fill

Number of days required to fill position.

Time To Join

Number of days required for candidate onboarding.

4.5 Candidate Dimension

Table Name

dim_candidate

Purpose

Store candidate master information.

Attributes

candidate_id

candidate_name
age
gender
education
experience_years

4.6 Department Dimension

Table Name

dim_department

Purpose

Store hiring department information.

Attributes

department_id
department_name

4.7 Recruiter Dimension

Table Name

dim_recruiter

Purpose

Store recruiter information.

Attributes

recruiter_id
recruiter_name

4.8 Location Dimension

Table Name

dim_location

Purpose

Store recruitment location information.

Attributes

location_id
location_name

4.9 Source Dimension

Table Name

dim_source

Purpose

Store candidate source information.

Attributes

source_id
source_name

4.10 Date Dimension

Table Name

dim_date

Purpose

Enable time intelligence analysis.

Attributes

date_key
full_date
year_num
quarter_num
month_num
month_name
week_num
day_num
day_name

4.11 Relationship Mapping

Dimension	Fact
dim_candidate	fact_recruitment
dim_department	fact_recruitment
dim_recruiter	fact_recruitment
dim_location	fact_recruitment
dim_source	fact_recruitment
dim_date	fact_recruitment

Relationship Type

One To Many

Cross Filter Direction

Single Direction

4.12 Data Modeling Benefits

Faster Queries

Smaller dimensions reduce query complexity.

Better DAX Performance

Measures calculate faster.

Easier Dashboard Design

Business users understand model quickly.

Scalability

Additional dimensions can be added later.

CHAPTER 5 – POWER BI IMPLEMENTATION

5.1 Data Import Process

Step 1

Open Power BI Desktop

Step 2

Select

Get Data

Step 3

Select

MySQL Database

Step 4

Connect to database

HR_ANALYTICS

Step 5

Import Tables

fact_recruitment
dim_candidate
dim_department
dim_recruiter
dim_location

dim_source
dim_date

5.2 Data Model Validation

The following validations were performed.

Candidate Dimension

1 : Many

Department Dimension

1 : Many

Recruiter Dimension

1 : Many

Location Dimension

1 : Many

Source Dimension

1 : Many

Date Dimension

1 : Many

5.3 DAX Measures

Total Applications

Total Applications =
COUNTROWS(fact_recruitment)

Total Hires

```
Total Hires =
CALCULATE(
COUNTROWS(fact_recruitment),
fact_recruitment[joining_status]="Joined"
)
```

Total Offers

```
Total Offers =
COUNTROWS(
FILTER(
fact_recruitment,
fact_recruitment[offer_status] <> BLANK()
)
)
```

Offer Acceptance %

```
Offer Acceptance % =
DIVIDE(
[Total Hires],
[Total Offers]
)
```

Average Hiring Cost

```
Average Hiring Cost =
AVERAGE(fact_recruitment[hiring_cost])
```

Average Time To Fill

```
Average Time To Fill =
AVERAGE(fact_recruitment[time_to_fill_days])
```

Average Skill Score

```
Average Skill Score =
AVERAGE(fact_recruitment[skill_score])
```

Average Technical Score

```
Average Technical Score =
AVERAGE(fact_recruitment[technical_score])
```

Average Communication Score

Average Communication Score =
AVERAGE(fact_recruitment[communication_score])

Average Overall Rating

Average Overall Rating =
AVERAGE(fact_recruitment[overall_rating])

5.4 Calculated Columns

Age Group

Age Group =
SWITCH(
TRUE(),
fact_recruitment[age] <= 25,"18-25",
fact_recruitment[age] <= 30,"26-30",
fact_recruitment[age] <= 35,"31-35",
fact_recruitment[age] <= 40,"36-40",
"40+"
)

5.5 Dashboard Design Principles

The dashboard was designed using the following principles.

Simplicity

Clean layout with minimal clutter.

Consistency

Uniform color palette across pages.

Readability

Clear KPI presentation.

Interactivity

Cross-filtering enabled.

Scalability

Additional pages can be added later.

5.6 Theme Colors

Primary Blue

#0056D2

Success Green

#28A745

Warning Yellow

#FFC107

Danger Red

#DC3545

Dashboard Background

#F8F9FA

5.8 Power BI Features Utilized

KPI Cards

Monitor business metrics.

Slicers

Interactive filtering.

Conditional Formatting

Visual performance indicators.

DAX Measures

Dynamic calculations.

Tooltips

Additional contextual insights.

Combo Charts

Multiple metrics in single visual.

Analytics Pane

Average lines and trend analysis.

Cross Highlighting

Interactive exploration.

5.9 Business Value

The dashboard enables HR teams to:

• Improve recruitment planning

• Monitor hiring costs

• Increase recruiter productivity

• Track candidate quality

• Improve hiring efficiency


HR RECRUITMENT EXECUTIVE DASHBOARD
Recruitment Performance Overview
Data as on : 31 Dec 2024
FILTERS
Total Applications
2,000
Total Hires
560
Total Offers
745
Offer Acceptance %
75.17%
▲ 8.45% vs LY
Avg Hiring Cost
₹ 21,324
▲ 6.32% vs LY
Avg Time To Fill
27.4
▼ -3.21% vs LY
Year
2024
Quarter
All
Department
All
Recruiter
All
Location
All
Source
All
Hiring Trend by Month (Applications)
Jan 120
Feb 135
Mar 150
Apr 180
May 175
Jun 210
Jul 190
Aug 165
Sep 170
Oct 185
Nov 205
Dec 215
Recruitment Funnel
Applied 2,000 100%
Interview 1,250 62.50%
Offer 745 37.25%
Joined 560 28.00%
Department Wise Hires
IT 162
Data Science 128
HR 86
Finance 72
Sales 64
Marketing 48
Recruiter Performance
Priya Sharma 96
Arjun Mehta 88
Neha Verma 72
Rahul Singh 68
Karan Patil 56
Sneha Iyer 48
Candidate Source Analysis
LinkedIn 32%
Referral 20%
Naukri 18%
Indeed 15%
Website 15%
Location Wise Hiring
Bangalore 142
Chennai 128
Hyderabad 102
Pune 74
Mumbai 58
Delhi 40
Kolkata 16
Hiring Cost by Department (Average)
₹ (in Thousands)
IT ₹26.8K
Data Science ₹24.5K
Finance ₹22.1K
HR ₹19.8K
Sales ₹18.6K
Marketing ₹17.2K
Average Hiring Cost: ₹21.3K
Monthly Offer vs Joined Trend
Total Offers
Total Hires
Offer Acceptance %
Jan 55 39 71%
Feb 60 44 73%
Mar 70 52 74%
Apr 75 57 76%
May 65 47 72%
Jun 80 62 78%
Jul 70 53 75%
Aug 65 48 74%
Sep 60 44 73%
Oct 75 56 75%
Nov 85 65 76%
Dec 85 66 77%
LY – Last Year Comparison

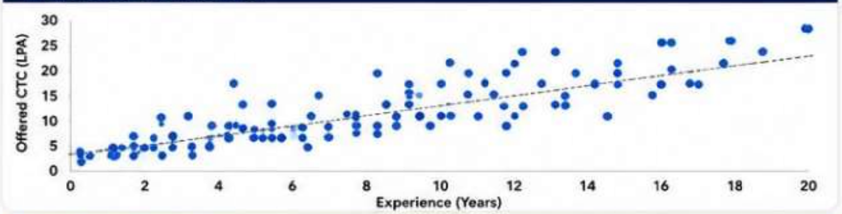
HR RECRUITMENT EXECUTIVE DASHBOARD - CANDIDATE QUALITY

Candidate Quality Overview

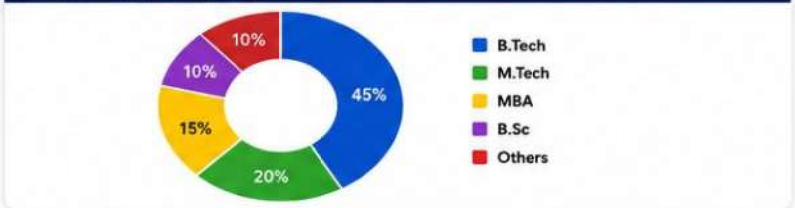
Data as on : 31 Dec 2024

Avg Skill Score	Avg Technical Score	Avg Communication Score	Avg Overall Rating	Avg Experience (Yrs)	Avg Offered CTC
78.6 / 100	76.8 / 100	74.2 / 100	82.4 / 100 ▲ 9.4% vs LY	6.2 Years ▲ 6.1% vs LY	10.45 LPA ▲ 8.7% vs LY

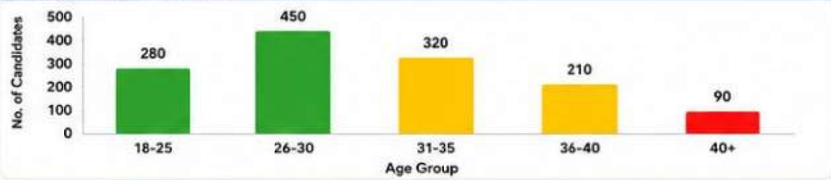
1. Experience vs Offered CTC



2. Education Distribution



3. Age Group Distribution



4. Department Wise Average Rating

5. Recruiter Wise Candidate Quality

6. Source Wise Candidate Quality

7. Skill vs Technical Score Comparison

8. Top 10 Highest Rated Candidates

	Candidate Name	Department	Recruiter	Skill Score	Technical Score	Comm. Score	Overall Rating (LPA)	Offered CTC (LPA)
1	Aarav Mehta	Data Science	Priya Sharma	92	95	90	95	18.5
2	Riya Nair	Data Science	Arjun Mehta	90	93	88	93	16.0
3	Karthik Reddy	IT	Priya Sharma	88	92	87	91	14.5
4	Neha Kapoor	Product	Neha Verma	85	90	86	90	13.0
5	Vikram Joshi	IT	Arjun Mehta	84	88	83	88	12.0
6	Sanika Kulkarni	Data Science	Priya Sharma	83	87	84	87	15.5
7	Aditya Singh	Product	Rahul Singh	82	86	83	86	11.5
8	Meghana Iyer	IT	Sneha Iyer	80	85	81	85	10.5
9	Tanya Sharma	Marketing	Neha Verma	79	83	80	83	9.5
10	Punit Agarwal	Finance	Karan Patil	78	82	78	82	8.5

All Scores are out of 100 | CTC in LPA

FILTERS

Department: All

Recruiter: All

Education: All

Source: All

Gender: All

Location: All

Clear All Filters

CHAPTER 6 – EXECUTIVE DASHBOARD OUTPUT ANALYSIS

6.1 Dashboard Overview

The Recruitment Executive Dashboard provides a centralized view of recruitment performance across departments, recruiters, locations, and candidate sources.

The dashboard enables management to monitor hiring efficiency, recruitment costs, offer conversions, and overall hiring performance.

The dashboard consists of:

- 6 KPI Cards
- 8 Interactive Visuals
- 6 Slicers
- Cross Filtering
- Conditional Formatting
- DAX Driven KPI

6.2 KPI Analysis

Total Applications

Displays the total number of applications received.

Business Value:

- Measures recruitment demand.
- Indicates talent acquisition reach.

Insight:

- Higher applications indicate stronger sourcing effectiveness.

Total Hires

Displays the number of candidates successfully joined.

Business Value:

- Measures recruitment success.

Insight:

- Helps track workforce growth.

Total Offers

Displays the number of offers released.

Business Value:

- Measures hiring pipeline strength.

Insight:

- Helps identify offer conversion trends.

Offer Acceptance %

Displays percentage of offers accepted.

Business Value:

- Measures attractiveness of compensation and employer brand.

Insight:

- Low acceptance indicates possible compensation issues.

Average Hiring Cost

Displays average hiring expenditure per candidate.

Business Value:

- Measures recruitment efficiency.

Insight:

- Helps control recruitment budget.

Average Time To Fill

Displays average days required to fill positions.

Business Value:

- Measures recruitment speed.

Insight:

- Lower values indicate efficient hiring.

6.3 Hiring Trend Analysis

Visual

Line Chart

Purpose

Analyze monthly application trends.

Business Insight

Management can identify:

- Peak hiring months
- Seasonal hiring patterns
- Recruitment demand fluctuations

6.4 Recruitment Funnel Analysis

Visual

Funnel Chart

Stages:

Applied
Interviewed
Offered
Joined

Purpose

Track candidate progression.

Business Insight

Identify bottlenecks in recruitment process.

Example:

If Interview stage drops significantly:

- Screening process may be ineffective.
- Candidate quality may be poor.

6.5 Department Wise Hiring Analysis

Visual

Clustered Bar Chart

Purpose

Compare hiring activity across departments.

Business Insight

Identify departments with:

- Highest hiring demand
- Workforce expansion
- Resource requirements

6.6 Recruiter Performance Analysis

Visual

Bar Chart

Purpose

Evaluate recruiter productivity.

Business Insight

Identify:

- Top performing recruiters
- Recruiters needing improvement
- Workload balancing opportunities

6.7 Source Effectiveness Analysis

Visual

Donut Chart

Purpose

Evaluate recruitment sources.

Sources:

- Referral
- LinkedIn
- Naukri

- Indeed
- Career Portal
- Consultancy

Business Insight

Identify highest ROI recruitment channels.

6.8 Location Wise Hiring Analysis

Visual

Column Chart

Purpose

Compare hiring activity across locations.

Business Insight

Identify:

- Talent demand by city
- Workforce distribution
- Regional expansion opportunities

6.9 Hiring Cost Analysis

Visual

Column Chart + Average Line

Purpose

Compare hiring costs by department.

Business Insight

Identify departments with:

- High hiring costs
- Budget overruns
- Cost optimization opportunities

6.10 Monthly Offers vs Joins Analysis

Visual

Combo Chart

Purpose

Compare offers released against actual joins.

Business Insight

Identify:

- Offer conversion rates
- Joining drop-offs
- Recruitment effectiveness

CHAPTER 7 – CANDIDATE QUALITY DASHBOARD OUTPUT ANALYSIS

7.1 Dashboard Overview

The Candidate Quality Dashboard evaluates candidate performance, recruiter effectiveness, compensation trends, and sourcing quality.

This dashboard focuses on quality rather than volume.

7.2 KPI Analysis

Average Skill Score

Measures candidate skill proficiency.

Average Technical Score

Measures technical competency.

Average Communication Score

Measures communication effectiveness.

Average Overall Rating

Measures overall candidate quality.

Average Experience Years

Measures average workforce experience.

Average Offered CTC

Measures compensation competitiveness.

7.3 Experience vs Offered CTC Analysis

Visual

Scatter Plot

Purpose

Analyze relationship between experience and compensation.

Insight

Typically:

Higher Experience → Higher Offered CTC

Outliers indicate premium candidates.

7.4 Education Distribution Analysis

Visual

Donut Chart

Purpose

Analyze candidate educational background.

Insight

Identify dominant qualification categories.

7.5 Age Group Analysis

Visual

Column Chart

Purpose

Analyze workforce demographics.

Insight

Helps understand hiring demographics.

7.6 Department Quality Analysis

Visual

Bar Chart

Purpose

Compare candidate quality across departments.

Insight

Identify departments attracting higher quality talent.

7.7 Recruiter Quality Analysis

Visual

Bar Chart

Purpose

Measure recruiter effectiveness.

Insight

Identify recruiters sourcing stronger candidates.

7.8 Source Quality Analysis

Visual

Column Chart

Purpose

Evaluate candidate quality by source.

Insight

Determine best recruitment channels.

7.9 Skill vs Technical Analysis

Visual

Combo Chart

Purpose

Compare skill and technical competency.

Insight

Identify skill gaps across departments.

7.10 Top Candidate Analysis

Visual

Conditional Formatting Table

Purpose

Highlight top performing candidates.

Insight

Supports talent identification and succession planning.

CHAPTER 8 – CONCLUSION

The HR Recruitment Analytics Dashboard successfully transformed raw recruitment data into meaningful business insights using MySQL and Power BI.

The project implemented:

- Star Schema Data Modeling
- Fact and Dimension Design
- SQL Data Mart
- Power BI Data Modeling
- DAX Measures
- KPI Monitoring
- Conditional Formatting
- Interactive Reporting

The solution enables HR teams to:

- Improve recruitment efficiency
- Reduce hiring costs
- Monitor recruiter performance
- Improve candidate quality
- Enhance workforce planning

The project demonstrates a complete end-to-end Business Intelligence implementation from data ingestion to executive-level reporting and provides a scalable foundation for future HR analytics initiatives.